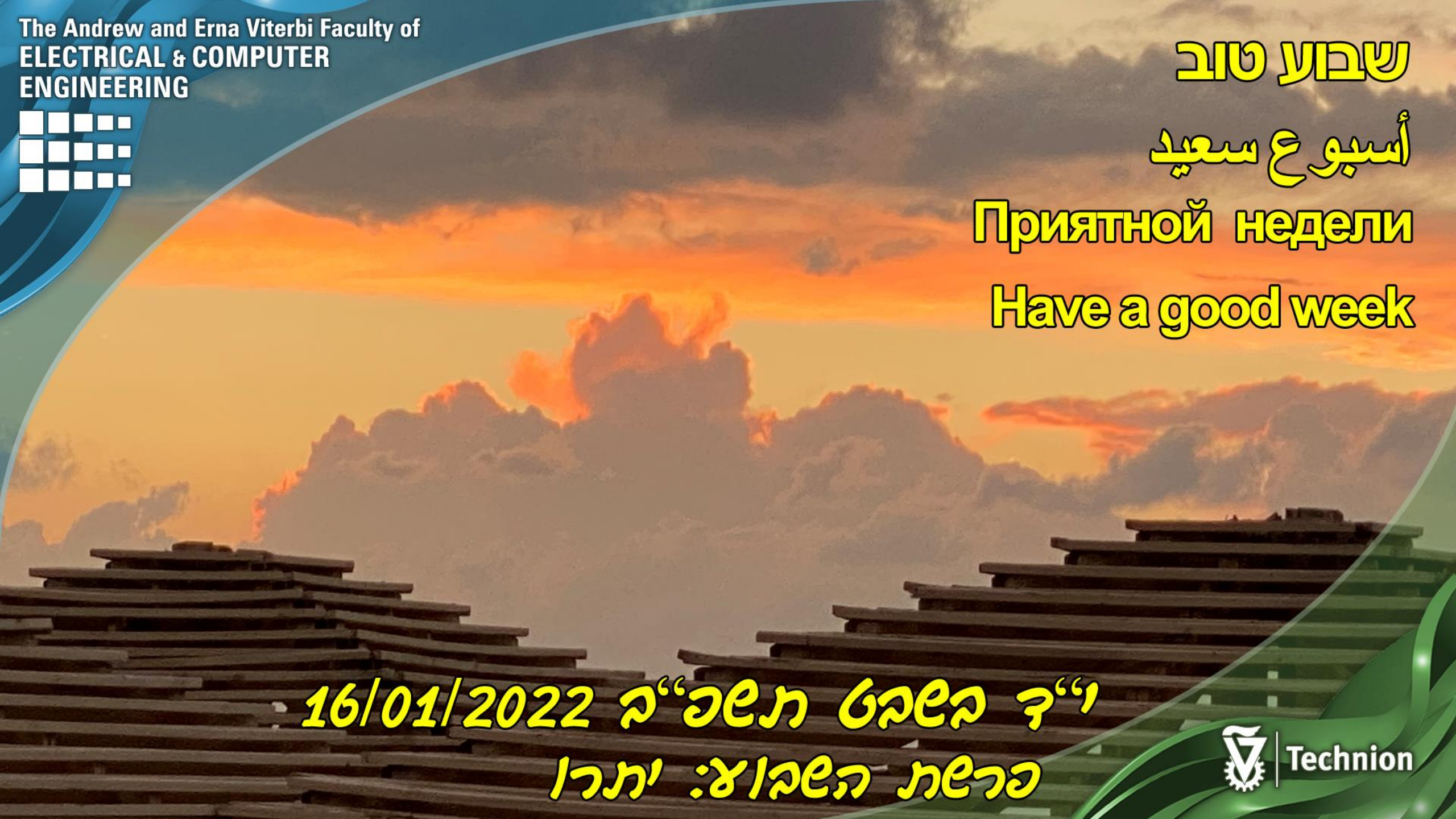

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ברוכים הבאים
Welcome
أهلًا وسهلًا
לפקולטה להנדסת חשמל ומחשבים
ע"ש אנדרו וארנה ויטרבי
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Have a good week
ו"ך בשבט תשפ"ב 16/01/2022
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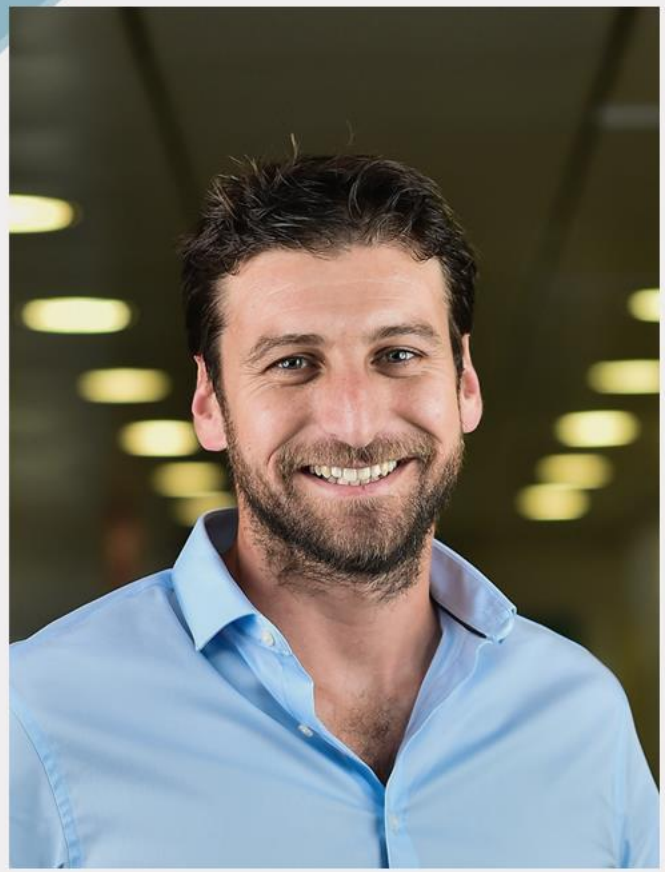
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ברכות

לד"ר אריאל אפשטיין

לקבלת הקביעות והעלייה

לדרגת פרופסור חבר





ינואר 2022

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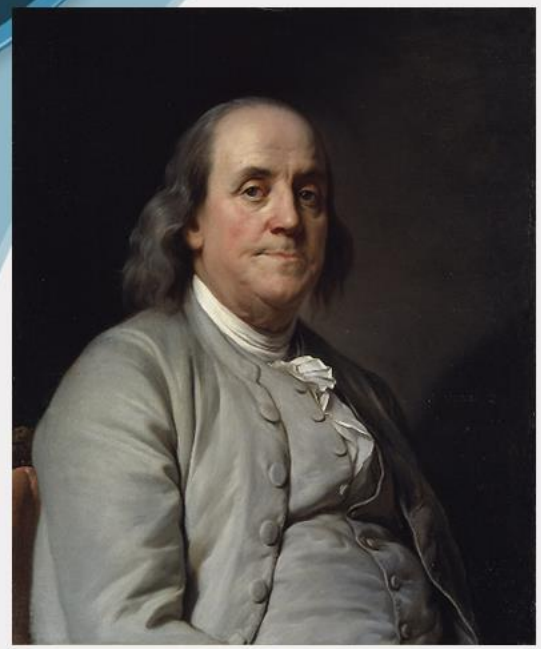
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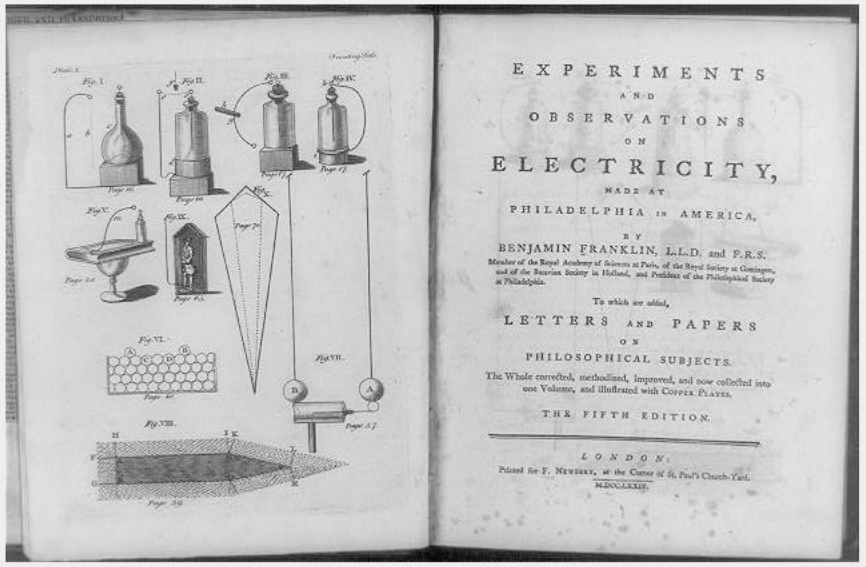
History of Science / Electricity

Benjamin Franklin (17, 1, 1706 – 17,4 , 1790)

One of the Founding Fathers of the United States and in many ways was "the First American"

As a scientist, he was a major figure for his discoveries and theories regarding electricity.









Benjamin Franklin saw the connection between the tiny electric sparks from a sulfur ball and the huge sparks of lightning – they were both flows of electric "fluid." He proved his point in his famous experiment where the moist cord from a kite flying in a thundercloud conveyed the electrical charge in the thundercloud to earth.

From Wikipedia, the free encyclopedia

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January 17, 2022

סמינר

Graduate Seminar

Electro-Optics and Microelectronics Seminar

Snap-Through bi-stability in carbon nanotube resonators



Sharon Rechnitz

PhD candidate
under the supervision
of Prof. Yuval Yaish

מרצה: **Sharon Rechnitz**

תאריך: **Monday 17/01/2022**

שעה: **14:30**

מקום: **Join Zoom Meeting**

In this research, we report a new class of CNT resonators in which the nanotube is buckled upward. This initial configuration enables a static out-of-plane movement with unique resonance frequency signature. We show that a small upward buckling yields record electrical frequency tunability, whereas larger buckling can achieve Euler-Bernoulli (EB) bi-stability, ...

הפקולטה להנדסת חשמל ומחשבים
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מועדון התעשייה

welcome to the club

KLA



Executive Partner

January 19, 2022

Machine Learning Seminar

סמינר

Precise characterization of the Distortion-Perception tradeoff



Dror Freirich

Ph.D. student
under the supervision of
Prof. Ron Meir

מרצה: Dror Freirich

תאריך: Wednesday 19/01/2022

שעה: 10:30

מקום: Join Zoom Meeting

The lower the distortion of an estimator, the more the distribution of its outputs generally deviates from the distribution of the signals it attempts to estimate. This phenomenon, known as the perception-distortion tradeoff, has captured significant attention in signal restoration, ...

Zoom link: https://technion.zoom.us/j/97829034114

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מכון טכנולוגי
לישראל

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ACRC

ברוכה הבאה



Apple

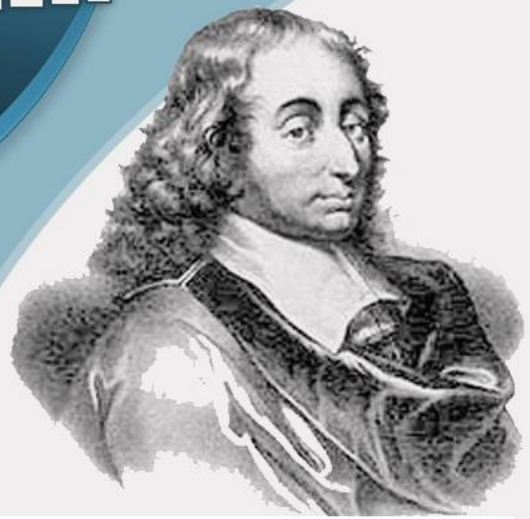
כשותפה בכירה ב ACRC

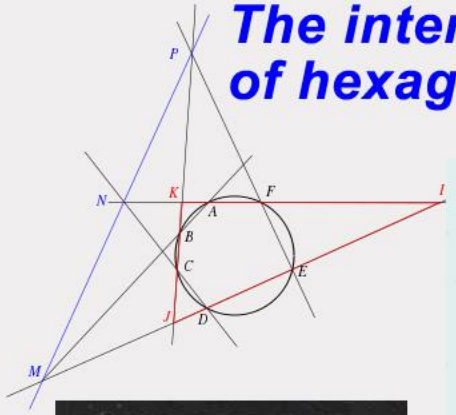
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History of Science

Blaise Pascal (19 June 1623 – 19 August 1662)

French mathematician, physicist, inventor, writer and Catholic philosopher



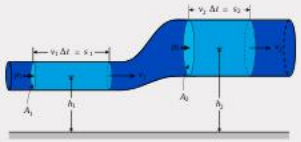


The intersections of the extended opposite sides of hexagon ABCDEF (right) lie on the Pascal line MNP (left)





A change in pressure at any point in an enclosed fluid at rest is transmitted undiminished to all points in the fluid

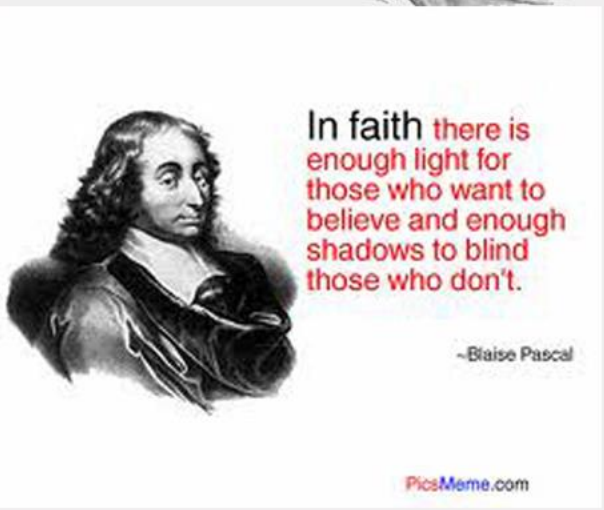


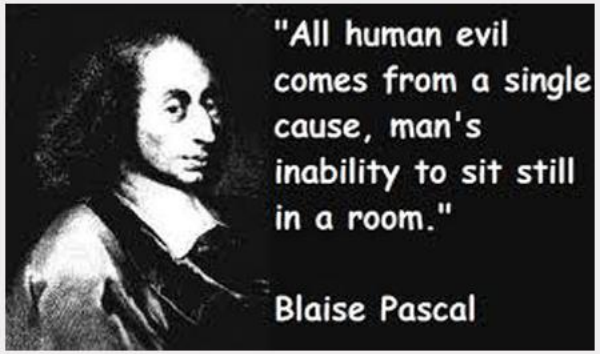
$$
\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 2 & & 1 & & & \\
 & & 1 & & 3 & & 3 & & 1 & & \\
 & 1 & & 4 & & 6 & & 4 & & 1 & \\
1 & & 5 & & 10 & & 10 & & 5 & & 1
\end{array}
$$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Each number in the triangle is the sum of the two directly above it.







From Wikipedia, the free encyclopedia

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SAVE THE DATE!

ACRC
Advanced Circuits Research Center

"Bringing ML to the extreme edge: a story of co-optimizing processor architectures, scheduling and models"

Abstract

Deep neural network inference comes with significant computational complexity, making their execution until recently only feasible on power-hungry server or GPU platforms. The recent trend towards real-time embedded neural network processing on edge and extreme edge devices requires a thorough cross-layer optimization. The talk will analyze what impacts NN execution energy and latency. Subsequently, we will present different research lines of Prof. Verhelst's lab exploiting and jointly optimizing NPU/TPU processor architectures, dataflow schedulers and conditional, quantized neural network models for minimum latency and maximum energy efficiency. This includes precision-scalable fully-digital designs, as well as compute-in-memory processors. Finally, this talk will make a case for more methodological design space exploration in the vast optimization space of embedded NN processors, using the ZigZag framework.

ACRC WEBINAR

"Bringing ML to the extreme edge: a story of co-optimizing processor architectures, scheduling and models"

Wednesday, January 19, 2022
10:00-11:30, online ZOOM session

Registration is free. Please register here:
https://acrc.net.technion.ac.il/registration-marian-verhelst/
Zoom link: to be provided after registration



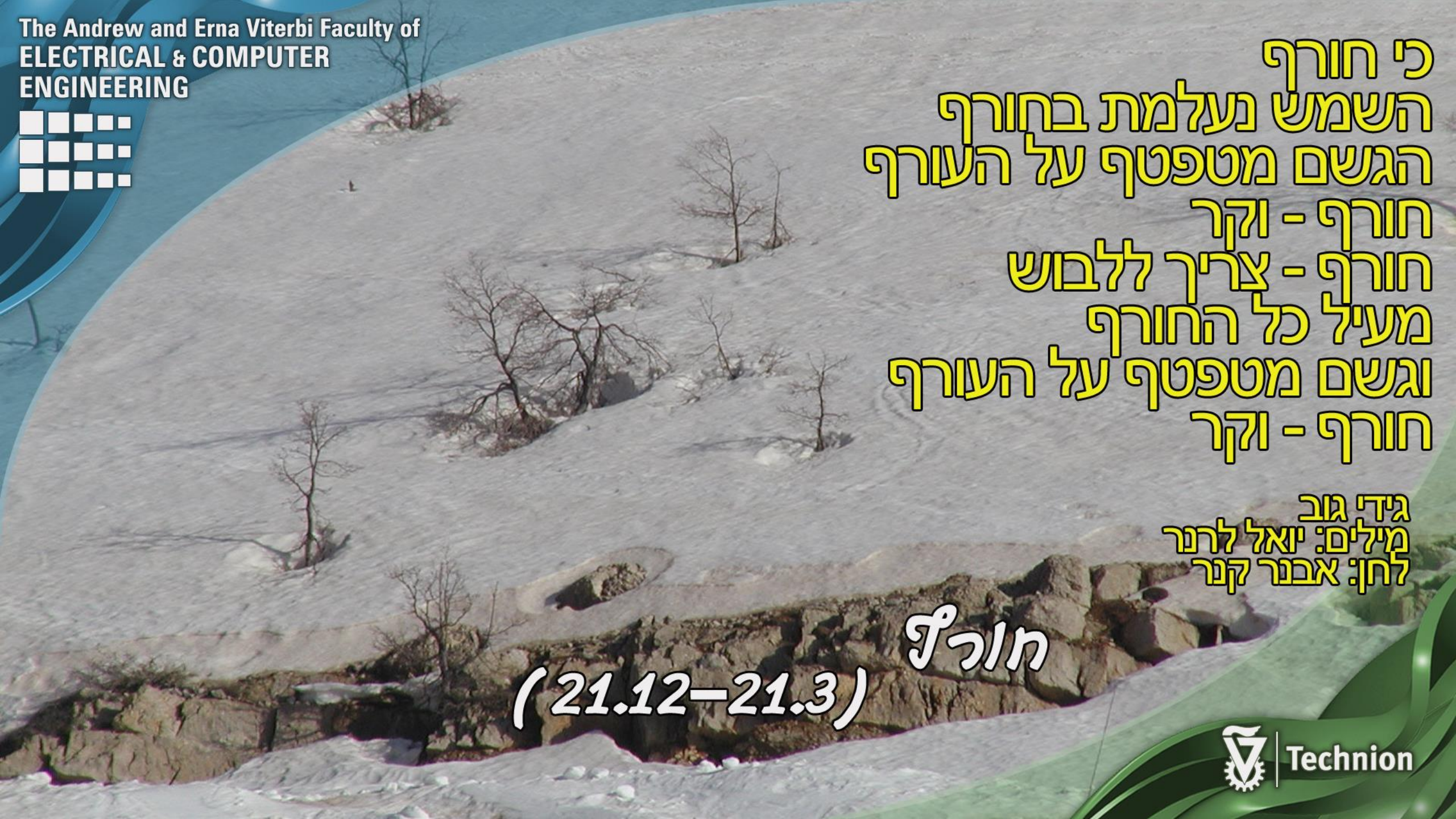
Prof. Marian Verhelst
KU Leuven, Belgium

KATHOLIEKE UNIVERSITEIT LEUVEN

Wednesday, January 19, 2022 at 10:00 (I D T)

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